

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO2 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
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Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A - APPLICATION BASED QUESTIONS

Q1.

Q2.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Solutions for Q71 and Q90

Q71

A smart agriculture system collects soil moisture readings from sensors placed randomly and needs periodic sorting for irrigation planning.

a) Selection Sort: Find the minimum element in the unsorted part and place it at the beginning. Repeat until sorted. Bubble Sort: Repeatedly compare adjacent elements and swap if out of order until no swaps occur.

b) Comparison:

- Selection Sort: Comparisons = $O(n^2)$, Swaps = $O(n)$.
- Bubble Sort: Comparisons = $O(n^2)$ (best case $O(n)$ with optimization), Swaps = $O(n^2)$.
- c) Recommendation: Selection Sort is more suitable because it performs far fewer swaps, making it efficient when write operations are costly.

Q90

A drone navigation system determines object boundaries using brute-force convex hull computation.

a) Brute-force Convex Hull:

1. Consider every pair of points.
2. Check whether all remaining points lie on the same side of the line formed by the pair.
3. If true, the line segment is part of the convex hull.
4. Repeat for all pairs to obtain the hull.

b) Time Complexity:

There are $O(n^2)$ pairs and each pair is checked against $O(n)$ points.

Overall time complexity = $O(n^3)$.

c) Why not scalable:

- Execution time increases rapidly for large datasets.
- High computational cost and power consumption.
- Not suitable for real-time drone navigation.
- Efficient algorithms such as Graham Scan or QuickHull ($O(n \log n)$ average) are preferred.